

Hometask - 05

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 Subject Name: Advanced Database
 Management System

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[A.]

A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed or may expire without completion.

For each registered user, calculate the confirmation rate, which is defined as:

$$\text{Confirmation Rate} = \frac{\text{Number of Successful Confirmations}}{\text{Total Confirmation Requests}}$$

If a user has never received any confirmation request, their confirmation rate should be considered **0**.

Display the user ID along with the confirmation rate rounded to **2 decimal places**. The result can be returned in any order.

Signups		Confirmations		
<u>user_id</u>	<u>time_stamp</u>	<u>user_id</u>	<u>time_stamp</u>	<u>action</u>
3	2020-03-21 10:16:13	3	2021-01-06 03:30:46	timeout
7	2020-01-04 13:57:59	3	2021-07-14 14:00:00	timeout
2	2020-07-29 23:09:44	7	2021-06-12 11:57:29	confirmed
6	2020-12-09 10:39:37	7	2021-06-13 12:58:28	confirmed
		7	2021-06-14 13:59:27	confirmed
		2	2021-01-22 00:00:00	confirmed
		2	2021-02-28 23:59:59	timeout

Output

<u>user_id</u>	confirmation_rate
2	0.50
3	0.00
6	0.00
7	1.00

Ans.

SELECT

s.user_id,

ROUND(

COALESCE(

AVG(CASE

WHEN c.action = 'confirmed' THEN 1.0

ELSE 0.0

END),

0

),

2

) AS confirmation_rate

FROM Signups s

LEFT JOIN Confirmations c

ON s.user_id = c.user_id

GROUP BY s.user_id;

[B.]

A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted from using the platform,

while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants.

For each day between **2013-10-01** and **2013-10-03**, determine the proportion of canceled trips among all valid trips. A trip should be considered valid only when both the customer and the driver associated with that trip are active users. The final result should display the date and the calculated cancellation rate rounded to two decimal places.

			Trips			
			id	client_id	driver_id	city_id
Users			status	request_at		
users_id	banned	role				
1	No	client	1	1	10	1
2	Yes	client	2	2	11	1
3	No	client	3	3	12	6
4	No	client	4	4	13	6
10	No	driver	5	1	10	1
11	No	driver	6	2	11	6
12	No	driver	7	3	12	6
13	No	driver	8	2	12	12
			9	3	10	12
			10	4	13	12

Output

Day	Cancellation Rate
2013-10-01	0.33
2013-10-02	0.00
2013-10-03	0.50

Ans.

SELECT

t.request_at AS day,

ROUND(

AVG(

CASE

WHEN t.status LIKE 'cancelled%' THEN 1.0

ELSE 0.0

END

),

2

```
) AS cancellation_rate  
FROM Trips t  
JOIN Users client  
    ON t.client_id = client.users_id  
JOIN Users driver  
    ON t.driver_id = driver.users_id  
WHERE client.banned = 'No'  
    AND driver.banned = 'No'  
    AND t.request_at BETWEEN '2013-10-01' AND '2013-10-03'  
GROUP BY t.request_at  
ORDER BY t.request_at;
```